Q. Implement following:

1. **Task to be done:**

   Implify the following query:

   SELECT ENAME, PNAME FROM EMP, ASG, PROJ

   WHERE (DUR > 12 OR RESP = "Analyst") AND EMP.ENO = ASG.ENO

   AND (TITLE = "Elect. Eng." OR ASG.PNO < "P3") AND (DUR > 12 OR RESP NOT= "Analyst")

   AND ASG.PNO = PROJ.PNO

2. **Objective:** Simplify the following query: SELECT ENAME, PNAME FROM EMP, ASG, PROJ WHERE (DUR > 12 OR RESP = "Analyst") AND EMP.ENO = ASG.ENO AND (TITLE = "Elect. Eng." OR ASG.PNO

   < "P3") AND (DUR > 12 OR RESP NOT= "Analyst") AND ASG.PNO = PROJ.PNO where select and project operations are applied as soon as possible to reduce the size of intermediate relations.

3. Code for experiment/practical:

COMPANY TABLE

Query 1 | company

Limit to 1000 rows

```
1 • select * from company
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

company_code	founder
C1	Monika
C2	Samantha

EMPLOYEE TABLE

Limit to 1000 rows

```
1 • select * from employee
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

employee_code	MANAGER_CODE	SENIOR_MANAGER_CODE	lead_manager	COMPANY_CODE
E1	M1	SM1	LM1	C1
E1	M1	SM1	LM1	C1
E2	M1	SM1	LM1	C1
E3	M2	SM3	LM2	C2
E4	M3	SM3	LM2	C2

LEAD_MANAGER TABLE

```
select * from lead_manager
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Lead_Manager_code	company_code
LM1	C1
LM2	C2

MANAGER TABLE

```
select * from MANAGER_TABLE
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

manager_code	senior_manager_code	lead_manager_code	company_code
M1	SM1	LMI	C1
M2	SM3	LM2	C2
M3	SM3	LM2	C2

SENIOR_MANAGER TABLE

```
1 •  select * from SENIOR_MANAGER
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

senior_manager_code	lead_manager_code	company_code
SM1	LM1	C1
SM2	LM1	C1
SM3	LM2	C2

4. Result/Output/Writing Summary:

FINAL OUTPUT: -

Query 1 | company

```sql
SELECT c.company_code, c.founder,
    (SELECT COUNT(DISTINCT Lead_manager_code) FROM Lead_manager WHERE company_code = c.company_code) AS total_lead_manager,
    (SELECT COUNT(DISTINCT senior_manager_code) FROM senior_manager WHERE company_code = c.company_code) AS total_senior_managers,
    (SELECT COUNT(DISTINCT manager_code) FROM Manager_table WHERE company_code = c.company_code) AS total_managers,
    (SELECT COUNT(DISTINCT employee_code) FROM Employee WHERE company_code = c.company_code) AS total_employees
FROM company c
ORDER BY c.company_code ASC
LIMIT 0, 1000;
```

Result Grid

company_code	founder	total_lead_manager	total_senior_managers	total_managers	total_employees
C1	Monika	1	2	1	2
C2	Samantha	1	1	2	2

Learning outcomes (What I have learnt):

1. Learned joining Data from Multiple Tables.
2. Learn how to sort query results using ORDER BY for clearer, organized output.
3. Learned about COUNT() and DISTINCT()